

Assignment : A+- Troubleshooting and helpdesk

Day 4: Component-Specific Troubleshooting Practical (1 hour): Identify and fix issues in faulty components (RAM, GPU, storage). Task: Create a checklist for diagnosing hardware issues.

TASK : 1

Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.



Practical status: Simulated procedure. The RAM reseating procedure was studied theoretically rather than physically performed. The expected result after correctly reseating the RAM is that the system detects the memory and starts normally.

Reseating RAM means:

Remove the RAM module from its slot and install it back firmly in the correct slot.

Sometimes RAM may not make proper contact with the motherboard because of dust, improper installation, or loose connection. Reseating it can help solve problems such as:

- No display
- PC not booting
- POST beep errors
- Random crashes
- RAM not being detected

Basic procedure

1. Shut down the computer

Turn off the PC completely and disconnect the power cable.

2. Open the computer case

Remove the side panel of the desktop carefully.

3. Locate the RAM

RAM is installed in long **DIMM slots** on the motherboard.

4. Release the RAM

Push the locking clips on the sides of the RAM slot outward.

5. Remove the RAM

Hold the RAM module by its **edges** and carefully remove it.

Avoid touching the gold contacts unnecessarily.

6. Inspect the RAM and slot

Check for:

- Dust
- Visible damage
- Broken clips
- Dirty contacts

7. Reinsert the RAM

Align the notch on the RAM with the key in the slot and press the module down evenly until the clips lock into place.

8. Start the computer

Reconnect the power and turn on the PC.

9. Check the result

Check whether:

- The computer starts normally.
- Display appears.
- POST errors/beeps disappear.
- RAM is detected by the system.

TASK : 2

Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.



A faulty GPU (Graphics Processing Unit) can cause a problem like

- No display
- Black screen error
- Distorted Graphics
- Colored lines
- Flickering screen
- Games Crashes

GPU Troubleshooting Checklist

Step	What to check	What to do
Identify the Symptom	What display problem is occurring	Note whether display is flickering , distorted
Check Monitor and cable	HDMI /Display port and Monitor	Check any cable is damaged or with another Monitor
GPU connection	GPU is Seating in PCI e slot	Turn off PC nad check GPU is well seated
Check GPU Power	GPU Power connector	Make sure GPU 6 / 8 pin power connector is firmly connected
Check Drivers	Graphics Driver	All drivers are installed or reinstall
Check temperature	GPU Fan	Fan is rotating or damaged
Check the system	Display and Graphics Performance	Power on and check application

Easy troubleshooting flow

Display problem

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Check monitor & cable

↓

Check GPU seating

↓

Check GPU power connection

↓

Check/reinstall driver

↓

Check temperature & cooling

↓

Test GPU

GPU troubleshooting should begin with simple checks such as the monitor, cable, and physical connections. Driver, power, and cooling problems should then be checked before concluding that the GPU itself is faulty

TASK : 3

Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.



To understand what happens when an SSD or HDD is disconnected and then reconnected.

Procedure:

1. Shut down the computer and disconnect the power.
2. Disconnect the SATA data/power cable from an SSD or HDD. For an NVMe SSD, remove it from the M.2 slot using the proper procedure.
3. Start the computer and enter the BIOS/UEFI setup.
4. Check the Storage or Boot Device section.
5. The disconnected drive is expected to be missing from the list of detected storage devices.
6. If the drive contains the operating system, the PC may show an error such as **“No bootable device found”** or **“Operating System not found.”**
7. Shut down the computer and reconnect the storage drive.
8. Start the computer and enter BIOS/UEFI again.
9. The storage drive should now appear in the detected storage devices.
10. If the boot settings are correct, the operating system should start normally.

Expected Observation

Condition	Expected Result
Storage Disconnected	Drive not detected in UEFI/BIOS
OS drive disconnected	No boot device error
Storage reconnect	Drive detected
Correct boot configuration	OS Starts normally

Conclusion:

Disconnecting an SSD or HDD can cause the BIOS/UEFI to stop detecting the storage device. If it is the boot drive, the computer may not be able to start the operating system. Reconnecting the drive correctly can restore detection and normal booting.

TASK : 4

Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.



Task 4: Diagnosing a PC That Randomly Restarts

Problem:

The PC randomly restarts during normal use.

Component

Among RAM, GPU, and storage, I would check the **RAM first**.

Reason:

Faulty, loose, or improperly seated RAM can cause system instability, freezes, crashes, and random restarts. Storage problems are more commonly associated with boot or file-access problems, while GPU problems generally produce display or graphics-related symptoms.

Diagnostic Steps:

1. Turn off the PC and disconnect the power.
2. Open the computer case and locate the RAM modules.
3. Check whether the RAM is properly seated and inspect it for visible damage or dust.
4. Remove and carefully reseat the RAM module.
5. If multiple RAM modules are installed, test one module at a time.
6. If necessary, test the RAM in another compatible memory slot.
7. Start the computer and check whether the random restart problem continues.
8. If the problem remains, continue troubleshooting other components such as the GPU, storage, power supply, cooling system, and motherboard.

Conclusion:

RAM is a reasonable first component to check when a PC randomly restarts because loose or faulty memory can cause system instability. Reseating and testing the RAM modules individually can help identify a memory-related problem.